

# Python Practice Handout

## Functions

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### 1. What is a function

A function is a way to give a name to a chunk of code. That is about all there is to it. Once the code has a name, you can run it whenever you want, as many times as you want.

### 2. Defining a function

Start with `def`, short for define. Making a function is also called defining one. After `def` write the name, then empty brackets, then a colon.

The line underneath is tabbed across, the same as with an `if` statement or a `for` loop. Whatever is tabbed over belongs to the function.

```
def bark():  
    print("Ruff ruff")
```

### 3. Defining does not run it

Run that program and nothing appears, even though there is a `print` inside. Defining a function only gives the code a name. It does not run it.

Output:

*(nothing)*

### 4. Calling the function

To actually run the code, write the function name with brackets after it. This is called calling the function.

```
def bark():  
    print("Ruff ruff")  
  
bark()
```

Output:

```
Ruff ruff
```

### 5. Call it as often as you like

Each call runs the code again. Write the definition once and use it as many times as you need.

```
def bark():  
    print("Ruff ruff")  
  
bark()  
bark()  
bark()
```

Output:

```
Ruff ruff
Ruff ruff
Ruff ruff
```

## 6. More than one line inside

A function is not limited to a single line. Every tabbed line belongs to it and runs on every call.

```
def bark():
    print("Ruff ruff")
    print("I'm a dog")

bark()
```

Output:

```
Ruff ruff
I'm a dog
```

## 7. Functions and for loops

Put a function call inside a for loop and it runs on every pass. This one barks a hundred times.

```
def bark():
    print("Ruff ruff")

for x in range(100):
    bark()
```

Output (first three lines):

```
Ruff ruff
Ruff ruff
Ruff ruff
```

As programs grow, functions keep them tidy. You name a chunk of code once and reuse it wherever you need it.

## 8. Try it yourself

Make a function called hello that prints Hello Nick, then call it. Try this on your own before looking below.

```
def hello():
    print("Hello Nick")

hello()
```

Output:

```
Hello Nick
```

If nothing appears, check that you called the function. Forgetting the last line is the most common mistake here.